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The Islamia University of Bahawalpur

Department of Data Science

Program/Semester: BS Data Science

Subject: Object-Oriented Programming

Instructor: Dr. Akmal Khan

Maximum Marks: 10

Time: 10 minutes

Date: 13/04/2025

Paper Type: MCQS (circle the right choice. Cutting and overwriting is not allowed)

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Section 1: Main Topic

1. Which of the following correctly shows the method resolution order (MRO) in Python? (1 marks)

- A. Breadth-first search
B. C3 linearization
C. Last-visit rule
D. Depth-first search

Q1: A ☐ B ☐ C ☐ D ☐

2. If 'inheritance' allows one class to derive the properties from another, what does 'composition' allow? (1 marks)

- A. Building more complex types with simple parts
B. Overloading operators
C. One class to have multiple constructors
D. Privacy of data members

Q2: A ☐ B ☐ C ☐ D ☐

3. Which of the following statements about Python's encapsulation is true? (1 marks)

- A. Python uses underscore prefixes to indicate private attributes.
B. Python does not support encapsulation at all.
C. Encapsulation in Python is achieved only through public methods.
D. Python enforces strict access controls at the syntax level.

Q3: A ☐ B ☐ C ☐ D ☐

4. Which of the following is a feature of Python's object-oriented programming? (1 marks)

- A. Namespaces
B. Pointers
C. Macros
D. Multiple inheritance

Q4: A ☐ B ☐ C ☐ D ☐

5. How can you prevent a method from being overridden in a subclass in Python? (1 marks)

- A. Using the staticmethod decorator
B. Using the super() method
C. Using the abstractmethod decorator
D. Using the @final decorator from the typing module

Q5: A ☐ B ☐ C ☐ D ☐

6. What is the result of the following Python code snippet? class Sample: def __init__(self, x): self.x = x obj = Sample(10) obj.x = 20 print(obj.x) (1 marks)

- A. Error
B. 10
C. 20
D. None

Q6: A ☐ B ☐ C ☐ D ☐

7. What is the purpose of the '__init__' method in Python classes? (1 marks)

- A. It is used to execute a block of code repeatedly.
B. It is used to delete an object.
C. It serves as a destructor for the class.
D. It initializes the object's properties when a new object is instantiated.

Q7: A ☐ B ☐ C ☐ D ☐

8. Which keyword is used to inherit a class in Python? (1 marks)

- A. extends
B. super
C. class
D. inherits

Q8: A ☐ B ☐ C ☐ D ☐

9. Which concept refers to using a shared interface method that can be implemented in various ways in different classes? (1 marks)

- A. Encapsulation
B. Abstraction
C. Inheritance
D. Polymorphism

Q9: A ☐ B ☐ C ☐ D ☐

10. What does the following code snippet illustrate? class A: pass class B(A): pass (1 marks)

- A. Multiple inheritance
B. Multi-level inheritance
C. Hierarchical inheritance
D. Single inheritance

Q10: A ☐ B ☐ C ☐ D ☐

Examiner's Signature:

Student's Signature:
